Precept 4: Warm Up Exercise

September 29, 2026

Problem

Consider the following two matrices:

$$R = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Let $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

1. Compute $R\vec{e}_2$ and $S\vec{e}_2$. How are they related? Now compute $R\vec{e}_1$ and $S\vec{e}_1$. What changed? Now compute $R^2\vec{e}_2$ and $S^2\vec{e}_2$. What would happen if we kept going (i.e. $R^n\vec{e}_2$ and $S^n\vec{e}_2$ for arbitrary n)?

2. What are the eigenvectors of R? What are the eigenvectors of S? Try to answer geometrically first. For context, an eigenvector is a vector $\vec{x}$ such that $A\vec{x} = \lambda\vec{x}$

3. Let's think about the singular value decomposition (SVD): $A = U\Sigma V^\top$. We will never learn how to compute this, so let's just think about it.

   (a) What would the SVD look like for R? **Hint:** What type of matrix is R?

   (b) Imagine feeding every point on the unit circle into S. What shape comes out? What do the singular values $\sigma_1 \geq \sigma_2$ correspond to in that picture?

   (c) In words, describe what the SVD for S would look like.

   (d) Take a guess at what the determinant of S is.

Solution

Question 1

Computing the $R\vec{e}_2$ and $S\vec{e}_2$ gives us

$$R\vec{e}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad S\vec{e}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \sqrt{2}\, R\vec{e}_2$$

So $S\vec{e}_2$ points in exactly the same direction as $R\vec{e}_2$; it's just $\sqrt{2}$ times longer. If we only ever looked at $\vec{e}_2$, we might think that S is a scaled version of R. But this is just a coincidence! Looking at $R\vec{e}_1$ and $S\vec{e}_1$, we have

$$R\vec{e}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad S\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \vec{e}_1$$

By now, it should be clear that R is a rotation of 45° clockwise. However, S didn't rotate $\vec{e}_1$, it just outputted $\vec{e}_1$ again. What's going on with S?

Applying the matrices again to $\vec{e}_2$ we find

$$R^2\vec{e}_2 = \begin{bmatrix}1\\0\end{bmatrix}, \qquad S^2\vec{e}_2 = S\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}$$

R has now turned $\vec{e}_2$ by a total of 90°. But $S^2\vec{e}_2$ is only about 63° from vertical, not 90°. In general,

$$S^n = \begin{bmatrix}1 & n\\0 & 1\end{bmatrix}, \qquad S^n\vec{e}_2 = \begin{bmatrix}n\\1\end{bmatrix}$$

If we keep applying R, the vector keeps going around the circle, with 8 applications taking us back to where we started $R^8 = I$. If we keep applying S, the vector just gets longer and longer, and it leans closer and closer to the horizontal axis without ever crossing it. The matrix S is what we call a *shear* matrix.

Question 2

Geometrically, an eigenvector is a vector that stays on its own line after being transformed. R turns every vector by 45°, so no vector stays on its own line. Hence R has *no* real eigenvectors. (Algebraically, the eigenvalues of R are complex, which we will ignore in this class.)

For S, we saw before that $\vec{e}_1$ did not get transformed. Hence,

$$S\begin{bmatrix}1\\0\end{bmatrix} = 1 \cdot \begin{bmatrix}1\\0\end{bmatrix}$$

and $\vec{e}_1$ is an eigenvector with eigenvalue 1. Is there another one?

For $\vec{v} = \begin{bmatrix}x & y\end{bmatrix}^\top$, the equation $S\vec{v} = \lambda\vec{v}$ gives us

$$S\vec{v} = \begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}x+y\\y\end{bmatrix} = \begin{bmatrix}\lambda x\\\lambda y\end{bmatrix} = \lambda\vec{v}$$

where we now have the following system of equations

$$x + y = \lambda x \qquad \text{and} \qquad y = \lambda y$$

Either $y = 0$ (giving $\vec{e}_1$ again), or $\lambda = 1$, which forces $x + y = x$ and hence $y = 0$, which contradicts $y \neq 0$. Therefore, S has exactly one eigenvector (up to scaling). This is also why repeated application of S pushed our vector toward the horizontal axis in Question 1.

Question 3

Part (a)

A rotation is already a rotation! So we can take $U = R$, $V^\top = I$, and $\Sigma = I$ to get

$$R = RII = U\Sigma V^\top$$

Notice that both singular values are 1. Since rotations preserve the norm of input vectors, no direction is scaled.

Part (b)

The unit circle gets mapped to a tilted ellipse. The singular values are the lengths of the semi-axes of the ellipse:

- σ_1 is how far the longest direction is stretched
- σ_2 is how much the shortest direction is squeezed

Also note that:

- The columns of U point along the axes of the ellipse
- For a rotation the ellipse is just the same circle, which is why $\sigma_1 = \sigma_2 = 1$.

Part (c)

As we did in **Part (b)**, imagine the input was a circle. Here is the procedure, reading $U\Sigma V^\top$ from right to left:

1. Rotate the circle clockwise (this is $V^\top$)
2. Stretch along the x axis by $\sigma_1 > 1$ and squeeze along the y axis by $\sigma_2 < 1$ to transform the circle into an ellipse (this is Σ)
3. Rotate again, counterclockwise, to tilt the ellipse into its final position (this is U)

Even though the shear looks nothing like a rotation, it can be built out of two rotations and a stretch!

For the curious, the exact singular values are

$$\sigma_1 = \frac{1+\sqrt{5}}{2} \approx 1.618, \qquad \sigma_2 = \frac{\sqrt{5}-1}{2} \approx 0.618$$

where σ_1 is the golden ratio, and the SVD looks like

$$S \approx \underbrace{\begin{bmatrix} 0.851 & -0.526 \\ 0.526 & 0.851 \end{bmatrix}}_{U\text{: rotate } 31.7^\circ \text{ counterclockwise}} \underbrace{\begin{bmatrix} 1.618 & 0 \\ 0 & 0.618 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 0.526 & 0.851 \\ -0.851 & 0.526 \end{bmatrix}}_{V^\top\text{: rotate } 58.3^\circ \text{ clockwise}}$$

Part (d)

Notice that

$$\sigma_1 > 1 > \sigma_2$$

which looks like

$$\sigma_1 \cdot \sigma_2 \approx 1$$

implying that

$$\det|S| = 1$$

In fact, it is indeed the case that the determinant of S is 1! We can show this via the exact singular values we noted above

$$\frac{1+\sqrt{5}}{2}\frac{\sqrt{5}-1}{2} = \frac{-1+\sqrt{5}-\sqrt{5}+5}{4} = \frac{4}{4} = 1$$

The determinant being equal to 1 implies that S is volume preserving.